

Comparing VR and Non-VR Driving Simulations: An Experimental User Study

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ABSTRACT

Up to now, most driving simulators use either small monitors or large immersive projection setups like 2D/3D screens or a CAVE. The recent improvements of VR-HMDs led to an increased application in driving simulation. However, the influence and comparability of various VR and non-VR displays has been hardly investigated.

We present results of a user study investigating the different influence of non-VR (2D, stereoscopic 3D) and VR (HMD) on physiological responses, simulation sickness, and driving performance within a single driving simulator. In the study, 94 participants performed the Lane Change Task.

Results indicate that a VR-HMD leads to similar data as stereoscopic 3D or 2D screens. We observed no significant difference regarding physiological responses or lane change performance. However, we measured significantly increased simulator sickness in the VR-HMD condition compared to stereoscopic 3D.

Keywords: Virtual Reality, Driving Simulation, Human Factors, Simulator Sickness.

Index Terms: H.5.1 [Information Interfaces and Presentation]: Multimedia Information Systems—Artificial, augmented, and virtual realities H.1.2 [Information Systems]: User/Machine Systems—Human factors

1 INTRODUCTION

Driving simulators have been an important tool for academia and research. Advantages of driving simulations compared to real driving experiments are safety, controllability and reproducibility [2]. Visualization of the environment is an essential part in driving simulation. In the past, large projection systems like 2D/3D screens or CAVEs have been widely used [4]. Due to recent technological advances of VR-HMDs, setups including these devices become increasingly popular (e.g. [7]).

However, the application of various display types in research pose a challenge: The question is, if results gathered in non-VR settings are comparable to results from a VR setup. In this area, some research has been done. Forster et al. compared 2D and stereoscopic 3D (S3D) simulations. They investigated lane keeping, distance estimation and simulator sickness. They found that S3D led to different (but not worse) lane keeping behavior and found no difference in distance estimation. However, in the S3D condition, simulator sickness was significantly higher [3]. Aykent et al. investigated the difference between 2D and VR-HMD regarding simulator sickness. They concluded that the VR-HMD induces significantly more simulator sickness but did not mention results about driving

performance [1]. To our knowledge, there has been no study comparing all of these conditions (2D, S3D, and VR-HMD) within one simulator and experiment.

The study presented in this paper aims to investigate how different display types (2D, S3D, and VR-HMD) influence simulator sickness, driving performance, and physiological responses of participants performing the Lane Change Task within a single driving simulator environment.

2 SYSTEM OVERVIEW

For driving simulation, we developed Car2Simulate with Unreal Engine 4. It connects to a Barco F50 120Hz WQXGA projector capable of 2D and active S3D. It illuminates a screen ($3.6m \times 2.25m$) in rear-projection mode (see fig. 1). In the stereo condition, we used volfoni EDGE RF glasses and a volfoni ActivHub RF-50. For the VR condition, an Oculus Rift DK2 (FOV: 100° (nominal)) was connected. (see fig. 2). In order to provide the feeling of “sitting in a car”, participants sat in a car mock-up (see fig. 3). It is equipped with a Thrustmaster TX Racing Wheel Leather Edition (incl. pedals), a Thrustmaster TH8A shifter, and a large-scale S3D spatial AR dashboard (see fig. 4). The final configuration in 2D and 3D provides a FOV of $60^\circ \times 35^\circ$. Simulation runs on a workstation equipped with an Intel Xeon CPU E5-2670 v3 with 2.3GHz, an NVIDIA Quadro M6000, and 64GB Ram. The driving simulation runs with 120Hz in 2D, 60Hz per eye in 3D, and 75Hz in VR with the DK2. A 5.2 audio system provides surround sound.

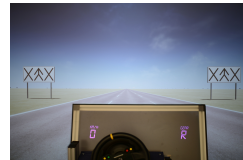


Figure 1: View in 2D & 3D

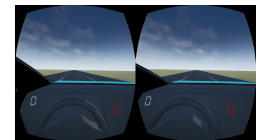


Figure 2: View in VR



Figure 3: Car mock-up



Figure 4: Dashboard concept

3 STUDY DESIGN, PROCEDURE AND SAMPLE

The cross-sectional laboratory study was conducted using a between-subject design. The between-subject factor was “condition” (2D, S3D, VR-HMD: Oculus Rift DK2). Dependent variables were simulator sickness, physiological responses and driving performance. Participants arrived, received a short oral briefing, filled out a demographic questionnaire, and were connected to

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the biofeedback system (PsyExpert Biofeedback 2000^{x-pert}). For the next approximately 10 minutes, they drove along a training track to get familiar with the simulator and changing lanes. During the final 20 minutes, they performed the Lane Change Task with 97 lane changes. After that, they were disconnected from the biofeedback system and filled out the Simulation Sickness Questionnaire. The sample consisted of $N = 94$ university students acquired via university mailing lists (74.5% male, 25.5% female; age: $M = 24.77, SD = 4.74$).

4 MEASURES

Driving performance was measured by correct lane changes in a shortened version of the Lane Change Task (LCT, [6]). Participants had to drive along a road with three lanes and had to change lanes according to street signs. A lane change is correct if the cars position before a sign confirms with the lane indicated by the previous sign. For physiological responses, we measured electromyography, heart rate variability (HRV), skin conductance, body temperature, blood volume pulse, 3D acceleration of the left and right hand, and respiration frequency. Simulator Sickness was measured according to the Simulator Sickness Questionnaire (SSQ, [5]).

5 RESULTS

5.1 Physiological Responses

For each measure, an analysis of variance (ANOVA) has been performed. Results show no significant differences between conditions. Analysis of the HRV's very low frequencies (VLF) results in $F(2,90)_{VLF} = 1.768, p_{VLF} = 0.176, \eta_p^2 = 0.038$. For all other measures applies $F < F_{VLF}, p > p_{VLF}$, and $\eta_p^2 < \eta_p^2_{VLF}$.

5.2 Simulator Sickness

Figure 5 shows the mean SSQ scores and standard deviations for each condition (2D: $M = 19.09, SD = 19.44$; S3D: $M = 13.49, SD = 13.05$; VR-HMD: $M = 30.91, SD = 28.24$). A one-way ANOVA indicates significant differences between conditions ($F(2,88) = 5.27, p = 0.007$). A post-hoc analysis (Bonferroni) revealed significant differences only between the S3D and VR-HMD condition ($SE = 5.54, p = 0.007, \eta_p^2 = 0.107$).

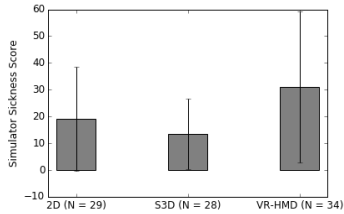


Figure 5: Mean SSQ score and standard deviation by condition.

5.3 LCT Performance

Figure 6 illustrates mean LCT scores and standard deviations for each condition (2D: $M = 96.66, SD = 1.06$; S3D: $M = 91.18, SD = 16.37$; VR-HMD: $M = 96.35, SD = 1.65$). A one-way ANOVA indicates significant differences between conditions ($F(2,91) = 3.06; p = 0.041$). However, a post-hoc Bonferroni analysis does not confirm this (2D - 3D: $SE = 2.42; p = 0.08$, 2D - VR-HMD: $SE = 2.27; p = 1.00$, 3D - VR-HMD: $SE = 2.29; p = 0.08$).

6 DISCUSSION

We detected no differences between conditions with respect to physiological responses. A reason for this might be that the rather relaxed nature of the LCT is not exciting enough for measurable

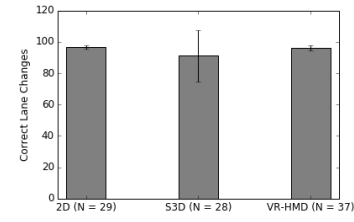


Figure 6: Mean LCT score and standard deviation by condition.

differences. Further, we observed no significant differences in driving performance. However, in a scenario where depth perception is more important (e.g. keeping distance to a leading vehicle), this result might change because of the depth cues offered by VR and S3D. Finally, compared to the S3D condition, the Oculus Rift DK2 induced significantly more simulator sickness. This difference is only based on self-reported data and is not backed by the physiological responses. We believe that the increased simulator sickness in the VR-HMD condition is because of the HMD's hardware limitations regarding tracking, resolution, and refresh rate. Limitations of the study are the not highly realistic vehicle physics of the simulation and the not up-to-date VR-HMD. Further, the sample consisted mostly of young students.

7 CONCLUSION AND FUTURE WORK

Results of the study indicate that, for simple tasks, display types do not influence driving performance and physiological responses significantly. Results from studies using these display types and simple tasks seem to be comparable. In order to generalize results, future work should include studies with more complex tasks. In addition to that, it seems to be worth investigating how participants perform while wearing a more up-to-date VR-HMD.

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